Edge-AI Driven Elastic Mesh PSO for Low-Altitude UAV IoT Networks in Zero-Infrastructure Environments

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Abstract. UAV swarms searching a disaster area without infrastructure must carry victim reports back to the responders over their own radio links. SENTINEL treats the swarm network as an elastic mesh that stretches and breaks during search and is re-formed only to deliver a confirmed report: Elastic-Mesh PSO (EM-PSO) plans waypoints, confirmed information triggers delivery, weak thermal detections get a second view, and an outage-based Signal-Spring keeps delivery links reliable. In about 13 000 simulated missions with a 2.4 GHz outage channel, 3–50 UAVs, six failure settings and seven tuned baselines, SENTINEL reports 23–29 % more victims within 60 s than the best baseline in urban rubble; PX4 software-in-the-loop runs agree with the simulator, and a mesh kept connected at all times reports far fewer victims in time.

Keywords. UAV swarm, edge AI, particle swarm optimization, search and rescue, intermittent connectivity, low-altitude IoT

Revised manuscript. Text in blue is new or changed in response to the reviews; see the response document for a point-by-point account.

1. Introduction

After an earthquake, small thermal-camera UAVs can sweep rubble and detect people on board [1,2]. A detection matters only once the responders receive it, over the UAVs' own short-range links, which buildings block, fading weakens and crashes cut. A connected swarm delivers at once but searches little, a spread-out one finds more but delivers by chance; we measure this discovery–delivery trade-off as the share of victims reported within 60 s.

Our first version held each UAV within 50 m of its nearest neighbour, but a fixed radius ignores rubble, a close neighbour does not imply a connected network (Lemma 1), and its 25 m test area was smaller than the radio range. This revision contributes (i) *information-triggered delivery*: a UAV stops searching when a victim it con-

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firmed has not reached the responder within a hold time T_h; (ii) *corroborative sensing*: weak detections get a lateral second view, fused so that a blocked angle cannot veto a clear one; and (iii) *EM-PSO with an outage-based Signal-Spring and link preservation*, active only during delivery, a safety property that keeps preserved links reliable (Proposition 1) and cuts link violations 37-fold; the reporting gains come from (i) and (ii). Code, generated scenarios, per-seed results and extended tables are available at `https://github.com/KS-KARTHIK-05/SENTINEL` (branch v2).

2. Related Work

Connectivity controllers keep the communication graph connected at all times, through its algebraic connectivity λ_2 [3,4] or edge potentials [5], with spanning trees reducing the links to keep [6]; all assume a disk radio model. Intermittent connectivity schedules meetings [7], and age-of-information collection serves sources at known positions [8], whereas victims are found by the search itself. Distributed PSO and Bayesian search [9, 10] assume communication when needed, and multi-agent reinforcement learning such as MAPPO [11] needs a trained policy. None combines online-discovered sources, a deadline and an obstruction-aware channel.

3. System Model

N UAVs fly at altitude h over tall buildings (no-fly, link-blocking) and low rubble that hides victims; a responder radio at the launch point is the only sink. Each slot Δt every UAV broadcasts its position, victim ledger and acknowledgements, received at j with SINR

$$\gamma_{ij} = \frac{P_t G^2\, 10^{-(\mathrm{PL}(d_{ij})+n_{ij}L_o+X_{ij})/10}\,|h_{ij}|^2}{\sigma^2 + \sum_{k\in\mathscr{I}_j} \bar{P}_{kj}}, \tag{1}$$

with log-distance path loss $\mathrm{PL}(d) = \mathrm{PL}(d_0) + 10\eta \log_{10}(d/d_0)$, n_{ij} obstacles at L_o each, correlated shadowing X_{ij}, Rician fading (Rayleigh when obstructed) and colliding transmitters $\mathscr{I}_j$. Decoding needs $\gamma_{ij} \geq \gamma_{\mathrm{th}}$, so

$$P_{\mathrm{out}}(d,n) = \mathbb{E}_X\left[F_{|h|^2}\left(\gamma_{\mathrm{th}}/(\bar{\gamma}(d,n)\,10^{X/10})\right)\right], \tag{2}$$

and the reliable range is $R_\varepsilon(n) = \max\{d : P_{\mathrm{out}}(d,n) \leq \varepsilon\}$. The link budget (2.4 GHz, $P_t = 20$ dBm, $G = 2$ dBi, $B = 20$ MHz, noise figure 7 dB, $\gamma_{\mathrm{th}} = 5$ dB) leaves 113 dB of path loss before fading margins, and profile parameters follow published ranges [12,13]. In urban rubble a link is reliable to 91 m in the open and 48 m behind one obstruction (Table 1), so the old 50 m radius describes one obstructed link. With τ_k the first confirmation of victim k and T_k the time its report reaches the responder (∞ if never), the team maximises

$$J = \frac{1}{|\mathscr{V}|}\sum_{k\in\mathscr{V}} e^{-T_k/\theta}, \qquad \theta = 300\,\mathrm{s}, \tag{3}$$

from local information.

Table 1. Channel profiles and reliable ranges at 5 % outage with no and one obstruction (K: line-of-sight Rician factor; L_o: loss per obstruction).

Profile	η	σ_X [dB]	K [dB]	L_o [dB]	$R_{0.05}(0)$ [m]	$R_{0.05}(1)$ [m]
open	2	2	10	6	2434	684
suburban	2.5	4	6	10	339	95
urban rubble	3	6	3	6	91	48
severe	3.5	8	Rayleigh	8	35	21

4. SENTINEL

Mode Switching. With $\mathscr{E}_i(t)$ the confirmed, undelivered entries UAV i owns (its own, and any older than $3T_h$, so a crashed UAV's copies still arrive), it leaves EXPLORE for DELIVER when

$$\max_{e\in\mathscr{E}_i(t)} (t-\tau_e) \geq T_h \tag{4}$$

and returns when $\mathscr{E}_i(t)$ is empty.

EM-PSO. Each planning period a UAV picks the waypoint in its reachable disk that maximises

$$F_i(x) = w_c\hat{G}_i(x) + w_d \sum_{e\in\mathscr{T}_i} c_e e^{-\frac{\|x-x_e\|^2}{2s^2}} + \frac{w_g(\|x_i-g_i\| - \|x-g_i\|) - w_t\|x-x_i\|}{vT_p} - S_i(x), \tag{5}$$

with $\hat{G}_i$ the expected new detections, $\mathscr{T}_i$ the tentative entries, g_i a frontier goal and w_t a travel cost. The Signal-Spring acts only while tethered (delivering or relaying):

$$S_i(x) = \lambda_i \sum_{j\in\mathscr{S}_i} \left(\frac{[P_{\text{out}}(\|x-x_j\|, n(x,x_j)) - \varepsilon_i]_+}{\varepsilon_i}\right)^2, \tag{6}$$

where $\lambda_i = \lambda_0(1+\beta U_i)$, $\varepsilon_i = \varepsilon/(1+\beta U_i)$ and U_i counts entries only i holds; it penalises outage, not distance. A constriction PSO ($\chi = 0.7298$, $c_1 = c_2 = 2.05$) [14] maximises Eq. (5).

Link Preservation. $\mathscr{S}_i$ holds i's neighbours in the local minimum spanning tree of reliable links [6].

Lemma 1. *Nearest-neighbour distance constraints do not imply connectivity.*

Proof. UAVs at $(0,0)$, $(10,0)$, $(200,0)$, $(210,0)$ with $R = 50$ m each have a neighbour at 10 m, yet no link joins the pairs. □

While tethered, UAV i scales its step u_i by the largest $\alpha \in [0,1]$ such that $x_i' = x_i + \alpha u_i$ satisfies, for all $j \in \mathscr{S}_i$,

$$\|x_i' - m_{ij}\| \leq \tfrac{1}{2}r_{ij} - \tfrac{1}{2}v\,a_{ij}, \qquad r_{ij} = R_\varepsilon\big(n^{\delta_{ij}}(x_i', \hat{x}_j)\big), \tag{7}$$

with $\hat{x}_j$ the last reported position of j (age a_{ij}), $m_{ij} = (x_i + \hat{x}_j)/2$, $\delta_{ij} = v(\Delta t + a_{ij})$ and n^δ counting obstacles inflated by δ.

Proposition 1. *If $j \in \mathscr{S}_i$, $i \in \mathscr{S}_j$, both UAVs are tethered, speeds are bounded by v, positions are at most a old and both steps satisfy Eq. (7), link (i, j) is ε-reliable at $t + \Delta t$.*

Proof sketch. Each midpoint estimate is within $va/2$ of the true one, so Eq. (7) and the triangle inequality give $\|x'_i - x'_j\| \le \frac{1}{2}(r_{ij} + r_{ji})$; the inflated count bounds the true one, and R_ε is non-increasing in it. □

Corroborative Sensing. An entry with $0.40 \le c_e < 0.70$ asks the nearest neighbour for a view from ± 15 m perpendicular to the line of sight, which at $h = 30$ m looks 26.6° off nadir past a slab edge at only 12 % more slant range. Keeping the best confidence $c_{e,v}$ per viewing sector v, SENTINEL fuses

$$\operatorname{logit} c_e = \max_v \operatorname{logit} c_{e,v} + \sum_{v \neq v^\star} [\operatorname{logit} c_{e,v}]_+, \tag{8}$$

with $v^\star$ the best view: weak views from blocked sides add nothing instead of vetoing it, as under naive-Bayes fusion.

5. Evaluation

Setup. One library runs in a fast simulator and in ROS 2 nodes flying PX4 quadrotors in Gazebo, both dropping packets by sampled SINR. Each run generates an area from its seed (25 victims, rubble, warm debris) and lasts 600 s; by default it is 400 m square, $N = 8$, urban rubble, Poisson crashes ($0.01\ \mathrm{s}^{-1}$), UAVs at 30 m and 5 m/s. The detector is a parameterised model of confidence versus range and occlusion, with false alarms from warm debris. Baselines: B0 PSO; B1 our first version; B2 PSO with an outage penalty; B3 λ_2 control with exact λ_2 [4]; B4 edge potentials [5]; B5 MAPPO [11]; B6 sweep search. All were tuned with one Optuna budget on J over separate seeds (SENTINEL: $T_h = 116$ s, $\lambda_0 = 1.20$, $\beta = 1.09$), and MAPPO was trained to a plateau. Seeds 0–29 are shared by all methods.

Results. SENTINEL leads in all four channel profiles, significantly against every baseline except in the severe one (open 0.577 vs. 0.501, suburban 0.560 vs. 0.456, urban 0.575 vs. 0.475, severe 0.433 vs. 0.377). In urban rubble it reports 23–29 % more victims than the best baseline in every failure setting (Table 2). On the 400 m area it beats every baseline significantly from 3 to 50 UAVs (Table 3); on a denser 200 m area it leads at every size but significantly only at $N = 50$ (0.809 vs. 0.779 for MAPPO).

Where the Gains Come From. Removing fusion (0.519) or delivery (0.523) lowers the urban share significantly from 0.575 (Table 4); a mesh kept connected at all times falls to 0.03–0.13. With SENTINEL's fusion rule the best urban baseline rises from 0.475 to 0.507 (B3) yet all seven stay significantly behind, while in a dense 50-UAV swarm baselines with fusion reach 0.815–0.865 and B4 leads significantly. With eight UAVs, $T_h = 15$ s raises the share over never delivering from 0.523 to 0.599 (urban) and 0.419 to 0.501 (severe, 90th-percentile latency 74 s → 41 s); with four UAVs the effect is not significant.

Table 2. Share reported within 60 s under six failure settings (urban rubble, 400 m, $N = 8$): none; one crash; 30 % of the swarm at once (simult.); three cascading crashes (casc.); Poisson crashes at twice the default rate; transmit-power brownout (brown.). Mean ± 95 % CI half-width over 30 seeds; best in bold; *: differs from SENTINEL ($p < 0.05$, paired Wilcoxon, Holm).

Method	none	single	simult.	casc.	Poisson	brown.
SENTINEL	**0.69±0.04**	**0.69±0.04**	**0.67±0.04**	**0.64±0.04**	**0.46±0.05**	**0.69±0.04**
B0 PSO	0.54±0.04*	0.52±0.04*	0.51±0.04*	0.51±0.04*	0.34±0.06*	0.54±0.04*
B1 first version	0.50±0.05*	0.50±0.05*	0.50±0.05*	0.49±0.05*	0.36±0.05*	0.51±0.05*
B2 comm.-aware PSO	0.56±0.03*	0.55±0.03*	0.52±0.03*	0.52±0.04*	0.34±0.03*	0.55±0.04*
B3 λ_2 control	0.53±0.03*	0.52±0.04*	0.51±0.04*	0.49±0.03*	0.34±0.05*	0.53±0.03*
B4 edge potentials	0.19±0.04*	0.16±0.03*	0.14±0.04*	0.16±0.04*	0.09±0.03*	0.18±0.04*
B5 MAPPO	0.50±0.04*	0.47±0.04*	0.45±0.04*	0.46±0.04*	0.33±0.05*	0.49±0.04*
B6 sweep	0.52±0.04*	0.51±0.04*	0.49±0.04*	0.49±0.04*	0.35±0.05*	0.52±0.04*

Table 3. Share reported within 60 s versus swarm size (400 m, urban rubble, Poisson failures). Mean ± 95 % CI half-width over 30 seeds; best in bold; *: differs from SENTINEL ($p < 0.05$, paired Wilcoxon, Holm).

Method	$N = 3$	$N = 5$	$N = 10$	$N = 20$	$N = 50$
SENTINEL	**0.20±0.04**	**0.39±0.06**	**0.67±0.04**	**0.75±0.03**	**0.79±0.03**
B0 PSO	0.15±0.03*	0.29±0.05*	0.53±0.04*	0.62±0.03*	0.66±0.04*
B1 first version	0.11±0.03*	0.23±0.05*	0.50±0.05*	0.65±0.03*	0.71±0.04*
B2 comm.-aware PSO	0.15±0.04*	0.31±0.05*	0.53±0.04*	0.64±0.04*	0.69±0.03*
B3 λ_2 control	0.14±0.04*	0.27±0.05*	0.53±0.04*	0.64±0.04*	0.67±0.03*
B4 edge potentials	0.02±0.01*	0.03±0.02*	0.24±0.05*	0.49±0.05*	0.71±0.04*
B5 MAPPO	0.15±0.04*	0.27±0.04*	0.50±0.05*	0.63±0.03*	0.73±0.03*
B6 sweep	0.15±0.04*	0.27±0.05*	0.44±0.04*	0.49±0.04*	0.62±0.03*

Table 4. Share reported within 60 s with SENTINEL components removed, and for the best baseline with and without SENTINEL's fusion rule (name in parentheses). Urban, severe: 400 m; dense: 200 m. Mean ± 95 % CI half-width over 30 seeds; best in bold; *: differs from SENTINEL ($p < 0.05$, paired Wilcoxon, Holm).

Variant	urban, N=8	severe, N=8	dense, N=50
SENTINEL	**0.57±0.05**	**0.43±0.05**	0.81±0.03
without fusion	0.52±0.05*	0.41±0.04	0.74±0.04*
without delivery (T_h=∞)	0.52±0.05*	0.42±0.05	–
without fusion, flank, delivery	0.52±0.05*	0.40±0.06	–
best baseline	0.47±0.04* (B3)	0.38±0.05 (B0)	0.78±0.04* (B5)
best baseline + fusion	0.51±0.04* (B3)	0.42±0.06 (B1)	**0.87±0.03*** (B4)

Link Preservation. Over all experiments, reliable preserved links failed within a step 65 times in 1 317 947 link-steps (4.9×10^{-5}). With the projection off on the same 4 settings and seeds, violations rose from 1 in 1 663 link-steps (in 1 of 120 runs) to 42 in 1 868 (in 21 of 120 runs), a 37-fold difference ($p < 10^{-4}$); short tethers leave the on-time share unchanged.

PX4/Gazebo and the First Scenario. 3 PX4 UAVs flew SENTINEL and B0 on 3 100 m worlds with one crash; reported victims, packet delivery and connected time agree with the fast simulator within 0.07; B0 is slightly ahead in both (0.60 vs. 0.53 in Gazebo). In

the 25 m, 3-UAV scenario of the first version all methods stay connected; confirmed victims are 0.71 for SENTINEL, 0.67 for B0 (not significantly fewer) and 0.63 for the first version (significantly fewer). Its earlier retention and discovery figures are withdrawn.

Detector. On HIT-UAV [1], YOLO26n [2] matches YOLOv8n in validation mAP50 (0.948 vs. 0.947) with 19 % fewer parameters and 31 % fewer operations, at 1.81 vs. 1.73 ms TensorRT latency on a desktop RTX 3060 (embedded latency not measured).

6. Limitations and Conclusion

Results are simulated, with published channel ranges and no flights; detector accuracy is a HIT-UAV validation result from above our 30 m, and the loop uses a hand-parameterised detector model. Victims within the 5 m association gate can merge, MAPPO used one reward design, and latency was measured on a desktop GPU only. SENTINEL reported more victims within 60 s than seven tuned baselines in most settings and was never significantly worse than any baseline in its original form. Fusion and information-triggered delivery produced the gains (in dense swarms fusion alone); the outage spring and link preservation are a safety property, cutting link violations 37-fold.

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